Scientific Data Analysis

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1 The model

We analyze a damped trigonometric function, defined for each new data point.

$$f(x) = e^{-0.1x}\sin(x). \tag{1}$$

2 Simulation in Julia

We define the function and evaluate a few points:

```
f(x) = exp(-0.1x) * sin(x)
x = 0:0.1:50
println("Points evaluated: ", length(x))
round.(f.(0:2:8), digits = 3)
```

```
Points evaluated: 501
5-element Vector{Float64}:
  0.0
  0.744
 -0.507
 -0.153
  0.445
```

The plot is generated separately, without showing the code behind it:

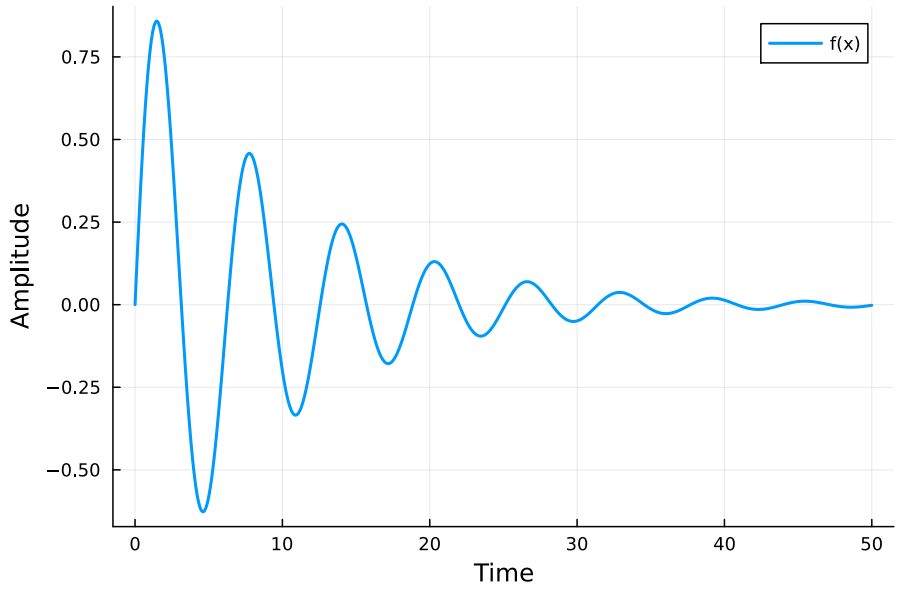


Figure 1: Damped function generated by Julia at compile time.

As figure 1 shows, the amplitude decays exponentially. The computed maximum is 0.8586.